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Mass-Anchored Glagov-Constrained CT-FFR: staged reduced-order model development in DISCHARGE

Project Description

Coronary CT-FFR is now routinely used to non-invasively triage stable chest pain, yet every current clinical implementation — including HeartFlow FFRCTA and Siemens cFFR — shares a common geometric blind spot: both derive the healthy reference vessel from the lumen itself and scale hyperaemic demand to vessel diameter rather than to myocardial tissue mass. These two simplifications introduce systematic errors precisely in the patients where accurate physiology matters most — those with positive arterial remodelling, left-ventricular hypertrophy, or prior myocardial infarction. This proposal addresses both errors within a single, transparent, reduced-order framework and tests them against invasive FFR in the DISCHARGE trial cohort. If validated, correcting these assumptions has the potential to reclassify a clinically meaningful fraction of borderline lesions, reducing unnecessary catheterisations in patients with positive remodelling and improving detection of physiologically significant disease in patients with abnormal ventricular mass. The primary deliverable is a validated, open-source CT-FFR pipeline with explicit reference geometry and physiologically grounded flow demand; DISCHARGE provides a dataset that no other cohort currently matches in combining outer-wall segmentations, an invasive FFR subset, long-term MACE follow-up, and a CT-accessible myocardial-mass workflow under calibration.

1 Starting point

1.1 State of the art

The two unresolved assumptions in clinical CT-FFR Invasive fractional flow reserve is an accepted physiological standard for lesion-level assessment and revascularisation guidance^{1–3}. CT-derived functional assessment extends this logic non-invasively and is incorporated into contemporary European and American diagnostic pathways as an adjunct to CCTA in selected patients^{4,5}. Despite this clinical progress, two geometric assumptions in every current commercial implementation remain untested at scale:

1. **The Glagov bias:** Coronary arteries remodel outward during early atherosclerosis, preserving lumen caliber despite substantial plaque burden^{6–8}. IVUS studies show that compensatory EEM expansion can coexist with substantial plaque burden while lumen area remains near-normal^{9,10}. Lumen-only pipelines — including HeartFlow FFRCTA and Siemens cFFR — infer the healthy reference vessel from the lumen itself, systematically underestimating reference size in positively remodelled segments. The consequence is that lesion severity and contraction-term pressure loss are underestimated in the patients most likely to benefit from better risk stratification.

2. **The flow-demand paradox:** Diameter-based flow scaling ($Q \propto D^{2.7}$) couples hyperaemic demand to vessel caliber. This creates a structural paradox in remodelled arteries — using the observed diameter underestimates demand; using the reconstructed reference diameter overclaims tissue burden. In patients with left-ventricular hypertrophy or prior infarction, the mismatch between vessel caliber and actual myocardial demand may drive systematic CT-FFR error. Demand is ultimately set by the mass of perfused myocardium, not by the vessel that serves it^{11–13}.

We hypothesise that outer-wall-constrained reference reconstruction and territory-mass-anchored hyperaemic boundary conditions will each improve diagnostic CT-FFR relative to lumen-only diameter-scaled baselines, and that the combined model will provide the largest gain. The proposal tests these as separate, independently attributable corrections within a pre-specified 2×2 design.

Methodological positioning The two assumptions targeted here are not unique to any one implementation: they are structural features of the current reduced-order CT-FFR paradigm. The table below maps existing methodological families against the two correction dimensions:

Approach	Outer-wall reference	Mass-anchored demand	Transparent / auditable	Runtime for UQ	Gap addressed here
3D CFD (Heart-Flow, Siemens cFFR)	No; lumen-only	No; diameter-scaled	No; opaque	Hours	Both
1D reduced-order (classical)	No; lumen-only	No; diameter-scaled	Yes	Seconds	Both
DL surrogate (ML-FFR)	No; lumen-only	No; diameter-scaled	No; opaque	Milliseconds	Both + interpretability
IVUS-guided CFD	Yes; IVUS outer wall	No; diameter-scaled	No; invasive	Hours	Flow demand

	Outer-wall Approach reference	Mass-anchored demand	Transparent / auditable	Runtime for UQ	Gap addressed here
MG-CT-FFR (this proposal)	Yes; CT outer wall (Glagov)	Yes; CT myocardial mass	Yes; open code	Seconds	Both

The most directly relevant recent prior art is Chen et al. (2024), who applied a distal-vessel CT-FFR framework for serial monitoring of epicardial disease¹⁴. That work illustrates the potential of CT-derived physiology beyond single-point diagnosis, but it still relies on lumen-derived reference geometry and diameter-based flow allocation. To our knowledge, no published CT-FFR study has combined outer-wall-constrained reference geometry with territory-mass flow allocation inside a transparent reduced-order framework. Transparent reduced-order modeling may also simplify later audit, documentation, and human oversight compared with opaque deep-learning surrogates.

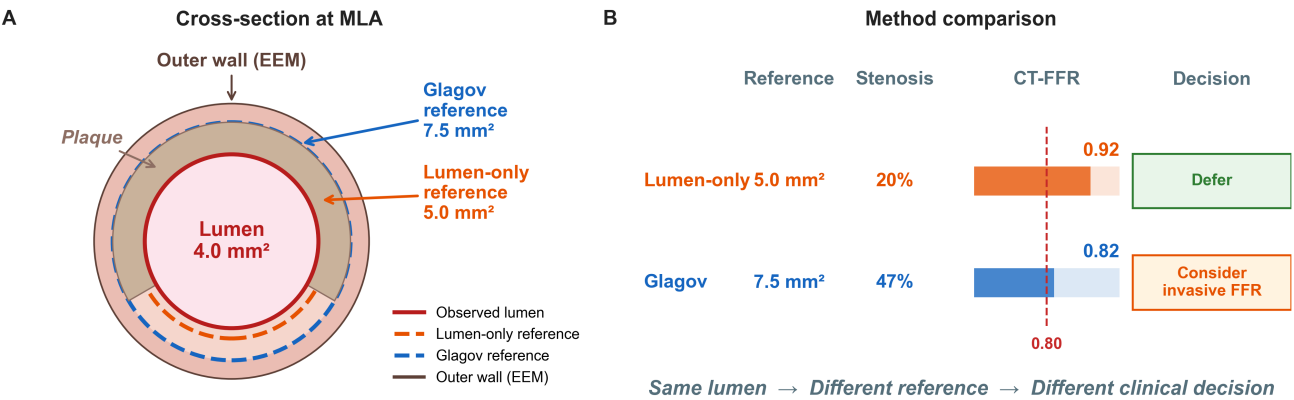


Figure 1: **The Glagov bias in CT-FFR (schematic + numerical demonstration).** (A) Schematic coronary cross-section at the minimum lumen area (MLA) showing the *same observed lumen* (4.0 mm²) but two different reference reconstructions: a lumen-only reference (dashed orange; 5.0 mm²) and a Glagov-constrained reference derived from the outer wall (dashed blue; 7.5 mm²). (B) Method comparison demonstrating that the smaller lumen-only reference underestimates stenosis severity (20%) and overestimates CT-FFR (0.92), suggesting no intervention, whereas the Glagov-constrained reference yields a higher stenosis estimate (47%) and lower CT-FFR (0.82), shifting the lesion into the borderline range relative to the clinical threshold (FFR = 0.80) and motivating consideration of invasive FFR. Numerical values are shown in-panel.

Clinical problem CT-FFR’s clinical value is highest precisely where existing models are most likely to err: intermediate stenoses, positively remodelled arteries, left-ventricular hypertrophy, and patients with prior infarction or scar. In each of these settings, both the reference geometry and the hyperaemic demand — the two inputs that determine pressure drop — are systematically misspecified by current approaches. An intermediate-severity lesion in a positively remodelled artery may be classified as

haemodynamically insignificant because the lumen-derived reference underestimates true reference diameter; the same lesion in a patient with LVH may be classified as non-obstructive because diameter-scaled demand underestimates actual myocardial oxygen requirement. These are not edge cases; they are the patients most likely to undergo unnecessary or deferred revascularisation on the basis of a CT-FFR report.

A reduced-order framework is well suited to this problem because it is fast enough to support Monte Carlo uncertainty analysis and contour-perturbation reproducibility testing, transparent enough for independent audit, and physically grounded in a way that allows each correction to be isolated, tested, and attributed separately. The DFG proposal frames this as a pre-specified 2×2 diagnostic development study, not as an immediate patient-outcome trial. Diagnostic accuracy against invasive FFR is the primary endpoint; exploratory prognostic analysis in the expanded cohort is a secondary, later phase.

Why DISCHARGE remains the right setting DISCHARGE is the only large-scale European coronary CT trial dataset that combines outer-wall segmentations from Medis QAngio CT, an invasive FFR subset, long-term MACE follow-up, and a CT-accessible myocardial-mass workflow under calibration in a single cohort. This combination is what makes the proposed 2×2 ablation study feasible: testing outer-wall correction requires outer-wall data; testing mass-anchored flow requires extractable myocardial mass; validating both against physiology requires an invasive reference standard. No other currently available dataset provides all four simultaneously in this exact staged-development configuration.

Concretely, the existing Medis repository contains **10,060 vessel segments across 1,684 patients**, with **709 patients** contributing overlapping dual-reader segmentations — a substantial geometry base from which to build. The invasive FFR subset ($n = 113$ before matching and quality-control filtering) provides the reference standard for diagnostic development. The broader DISCHARGE cohort with median 3.5-year MACE follow-up supports later exploratory prognostic analysis once the diagnostic model is validated. Segmentation coverage is not yet uniform across the full tree, and harmonization work is a named deliverable of WP1; this is a practical starting condition, not a disqualifying limitation.

1.2 Preliminary work

Existing vessel data Under the completed DFG Priority Programme SPP 2177 (grant 428223139), prior DISCHARGE work generated a substantial vessel-segmentation repository using Medis QAngio CT: 10,060 vessel segments across 1,684 patients, with 709 patients contributing overlapping dual-reader segmentations^{15,16}. The Medis platform has demonstrated high reproducibility for quantitative coronary measurements^{16,17}. Coverage is not yet a uniform proximal-to-distal coronary tree in all cases, and outer-wall quality is expected to vary in calcified segments; harmonization and geometry QC are therefore named deliverables of WP1. The project begins from a strong geometry seed, not from zero.

A representative DISCHARGE / Medis example is shown below.

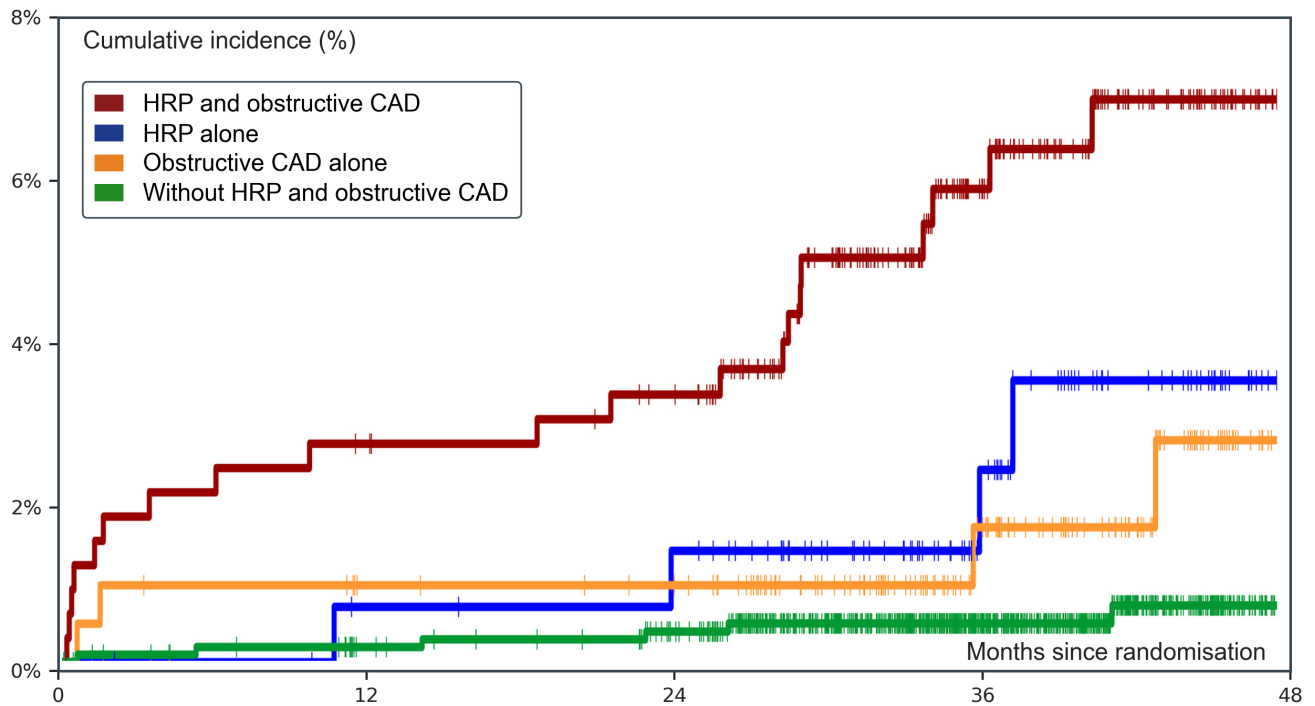


Figure 2: **Prognostic impact of high-risk plaque in DISCHARGE.** Time-to-event curves for major adverse cardiovascular events (MACE) in the CT arm of DISCHARGE. High-risk plaque features, alone or in combination with obstructive CAD, stratify risk over median 3.5-year follow-up and support later exploratory patient-level analyses in the harmonized CT cohort.

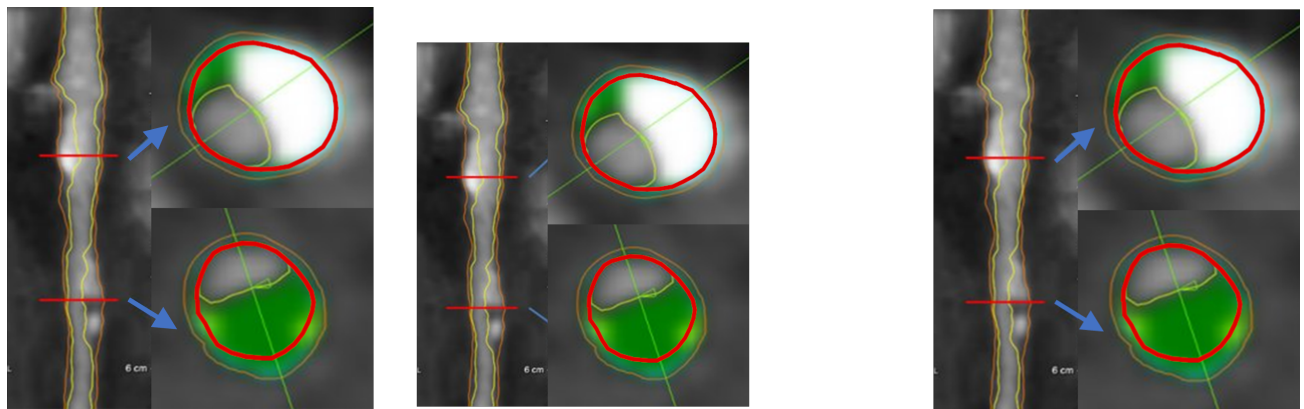


Figure 3: **Coronary CT cross-section with positive remodelling (DISCHARGE / Medis QAngio CT example).** Medis QAngio CT lumen (orange) and outer-wall (blue) segmentations along a straightened multiplanar reformation (MPR) and corresponding cross-sectional views at/around the minimum lumen area. The example illustrates preserved lumen despite substantial plaque burden (positive remodelling) and the availability of outer-wall information required for Glagov-constrained reference reconstruction and large-scale MG-CT-FFR re-analysis in DISCHARGE.

Existing myocardial-mass workflow To support mass-anchored flow estimation, the project builds on an automated 3D left-ventricular myocardium segmentation workflow based on nnU-Net and existing whole-heart CT segmentation infrastructure^{18,19}. Approximately 300 DISCHARGE cases have been manually refined for task-specific development in prior internal project work. That prior track reported held-out Dice and CT-to-CMR agreement figures compatible with mass-based development, but the current repository does not yet package those pilot metrics as an auditable result table. They should therefore be treated as provisional internal status claims pending WP1 re-verification on the QC-locked cohort, where the locked pilot calibration figures will be reported explicitly.

Before myocardial mass is used as a locked quantitative boundary-condition input, the CT-derived mass output will be calibrated against a paired CT-CMR subset ($n \approx 50$) to define the cohort-wide calibration map²⁰. The stage-1 default is zero-intercept slope-only Deming calibration unless the paired subset justifies freeing the intercept.

The applicant group brings complementary expertise in quantitative cardiac CT, including low-dose 4D myocardial CT perfusion, AI-enabled plaque quantification, and quantitative imaging guidance work from the QCI Study Group^{21–25}.

Staged cohort strategy The project is organized around three nested cohorts:

Cohort	Purpose
A. FFR-linked development cohort	Primary model development and invasive validation
B. Geometry/robustness cohort	Outer-wall, Glagov, and segmentation-method development
C. Exploratory prognostic cohort	Secondary patient-level association testing after harmonization

This staged design reflects the actual data situation: the primary development endpoint is diagnostic validation in the FFR-linked subset, and cohort expansion follows only after the pipeline is stable.

2 Objectives and work programme

2.1 Anticipated total duration of the project

24 months

This timeline includes segmentation harmonization work, development-phase model validation, and only then exploratory expansion to a larger CT cohort.

The schedule follows a staged complexity escalation from lumen-only benchmarking to Glagov-aware geometry and then to mass-based flow allocation. Existing DISCHARGE imaging and segmentation work removes data-acquisition latency, so the project time is spent on harmonization, quality control, reduced-order implementation, and validation rather than on primary data collection.

2.2 Objectives

We will build and validate a staged reduced-order CT-FFR framework in which reference geometry and flow allocation are tested independently within a pre-specified 2×2 design.

2.2.1 Aim 1: Establish a defensible development-phase CT-FFR baseline against invasive FFR

We will first construct an analyzable FFR-linked cohort from the 113 available invasive FFR cases by performing:

- CT-to-FFR matching;
- lesion-site mapping;
- vessel-continuity quality control;
- exclusion of segments that cannot support stable flow derivation.

For stage-1 validation, the immediate geometric target is not a full coronary tree but a continuous target-vessel span extending from a proximal reference region, across the lesion, to a distal lumen segment beyond the lesion and, where sensor-location metadata are available, approximating the invasive FFR sensor location. This geometry is sufficient for development-phase CT-FFR in the matched vessel, but it is not equivalent to full-tree CT-FFR or to a lesion-only pressure-drop analysis.

Target-vessel linkage will be defined prospectively: each invasive FFR measurement will be matched to the CTA vessel carrying the pressure-wire sensor distal to the interrogated stenosis. Where pull-back traces or angiographic annotations are available, the hemodynamically dominant lesion will be assigned to the CTA stenosis immediately proximal to the largest documented pressure step. Vessels with diffuse serial disease but no defensible focal lesion mapping will remain eligible for vessel-level validation, but will be excluded from lesion-localized contraction-term analyses.

The primary baseline model is **L-CT-FFR**: lumen-only reference geometry with diameter-based flow scaling. Given matching losses, lesion-site ambiguity, and strict geometry QC, we conservatively expect approximately **70–90 analyzable paired target-vessel comparisons** in the broader matched-vessel pool from the recorded 113-case subset. For planning purposes, the strict four-arm complete-case primary subset is expected to retain roughly **45–60 paired vessels** once direct Level-0 MLA wall support and an acceptable locked LV-mass state are both required. At the broader-pool scale, and assuming the invasive subset has a mid-range ischemic prevalence, the study is sized for development-phase detection of moderate paired discrimination gains on the order of nominal $\Delta\text{AUC} \approx 0.09\text{--}0.10$, but not for confirmatory claims about very small differences between neighboring reduced-order variants. Under Holm correction, the first adjusted comparison is tested at one-sided $\alpha_1 = 0.0167$, which raises the nominal detectable paired ΔAUC to approximately **0.11–0.12** even before accounting for any further reduction in the strict complete-case subset. The primary claim is therefore model ranking and interval estimation, not definitive proof of superiority margins below that range, and the final strict-primary threshold may be larger still once the realized primary-set yield is locked.

2.2.2 Aim 2: Test geometry and demand corrections separately and jointly We will compare four pre-specified models:

	Geometry	Flow	Model
	Lumen-only	Diameter-based	L-CT-FFR
	Lumen-only	Mass-based	M-CT-FFR
	Glagov reference	Diameter-based	G-CT-FFR
	Glagov reference	Mass-based	MG-CT-FFR

	Diameter-based flow ($Q \sim D^{2.7}$)	Mass-based flow ($Q \sim M^{1.0}$)
Lumen-only geometry	L-CT-FFR • Lumen segmentation only • Reference: lumen taper • Flow: $Q \sim D^{2.7}$	M-CT-FFR • Lumen segmentation only • Reference: lumen taper • Flow: $Q \sim M^{1.0}$
Glagov ref geometry	G-CT-FFR • Lumen + outer-wall CT • Reference: Glagov-constrained • Flow: $Q \sim D^{2.7}$	MG-CT-FFR • Lumen + outer-wall CT • Reference: Glagov-constrained • Flow: $Q \sim M^{1.0}$ combined model

Figure 4: **2×2 factorial design of CT-FFR models (Aim 2)**. Rows distinguish the geometric reconstruction strategy: lumen-only segmentation vs. Glagov-constrained outer-wall reference. Columns distinguish the primary flow-allocation law: diameter-based scaling vs. CT-derived myocardial-mass-based allocation. The four pre-specified models occupy one cell each, enabling separate assessment of geometry and flow-allocation effects and whether the combined model adds value beyond either component alone. **MG-CT-FFR** (highlighted) is the combined model, but advancement to Aim 3 follows the pre-specified ranking hierarchy: highest AUC, then lowest mean absolute error, then calibration closest to the identity line. Allometric sensitivity analyses varying the mass exponent over $\gamma \in [0.75, 1.0]$ are evaluated separately and are not additional cells in the core 2×2 comparison.

This 2×2 design is the cleanest way to quantify:

- the added value of Glagov-aware reference geometry;
- the added value of territory-mass flow allocation;
- whether the combination is additive.

The primary inferential set is the strict four-arm complete-case paired target-vessel subset whenever all required inputs are available. Maximal eligible pairwise subsets are retained only as sensitivity analyses. The inferential frame is development-phase **superiority-style comparison**, not non-inferiority: **L-CT-FFR** is the reference model, and **G-CT-FFR**, **M-CT-FFR**, and **MG-CT-FFR** are the three planned contrasts. The candidate advanced to Aim 3 will be selected by a pre-specified hierarchy: highest AUC first, then lowest mean absolute error, then calibration closest to the identity line.

2.2.3 Aim 3: Expand to exploratory patient-level clinical testing Once the geometric processing pipeline is stable, the pre-specified best-performing model will be evaluated in a broader harmonized CT subset for exploratory patient-level association with available clinical follow-up. The working exploratory endpoint will be **time-to-first major adverse cardiovascular event (MACE)** as adjudicated within DISCHARGE follow-up, analyzed first with the patient-level CT-FFR summary as a single covariate. This is a **secondary stage**, not the primary development endpoint, and not a decision criterion for the core model-development work.

Because processing completeness in the broader CT cohort depends on segmentation harmonization, vessel-level QC, and available outcome counts, this aim is intentionally not framed as a confirmatory prognostic study. Multivariable expansion will only be attempted if the final event count supports it, using the practical rule of approximately **10 outcome events per added covariate**; otherwise reporting will remain descriptive or limited to single-predictor Cox or logistic models according to the finalized follow-up structure.

2.2.4 Innovation statement Every current clinical CT-FFR implementation shares two assumptions that have never been tested at scale: that the healthy reference vessel can be inferred from the lumen, and that hyperaemic demand is proportional to vessel diameter. Both are likely wrong in the patients where CT-FFR clinical decisions are hardest. This proposal targets both in a single framework, with four features that distinguish it from prior CT-FFR development work:

1. **Outer-wall information as a reference-modeling target.** The Glagov outer-wall contour enters the model in the only term where it changes hemodynamics — the reference-dependent contraction loss — not as a plaque marker or secondary annotation. This makes the role of outer-wall data mechanistically explicit rather than empirically implicit.
2. **Territory-mass flow allocation.** Hyperaemic demand is estimated from the CT-derived mass of the perfused myocardial territory, decoupled from vessel caliber. This is the correct physiological frame for patients with LVH or prior infarction, where mass-to-caliber ratios are abnormal.
3. **Pre-specified 2×2 ablation.** The four-cell design with fixed parameters across all variants ensures that geometry and demand corrections are attributable independently and that the combination can be tested for additivity. No prior CT-FFR study has isolated these two corrections within the same cohort.
4. **Open, transparent, reproducibility-tested pipeline.** The reduced-order framework is fast enough to support Monte Carlo uncertainty quantification and contour-perturbation analysis in the same dataset used for validation. Code will be released openly, enabling external reproduction and adaptation — a standard that commercial CT-FFR platforms do not meet.

2.3 Work programme incl. proposed research methods

Shared notation The proposal uses the same core symbols as the companion manuscript and segmentation reference:

Symbol	Meaning
$A_{\text{lum}}(s)$	observed lumen cross-sectional area along vessel arc length
$A_{\text{outer}}(s)$ $A_{\text{ref}}(s)$	outer-wall cross-sectional area where available reconstructed local reference area profile, either lumen-envelope or Glagov-based
s	proximal-to-distal vessel arc length
M_{cal}	calibrated CT-derived LV myocardial mass entering the flow model
$M_{\text{ref,cal}}$	reference calibrated LV myocardial mass used to normalize the allometric sensitivity model (primary anchor 250 g; defined explicitly below)
f_v	vessel territory fraction
q_{rest}	resting perfusion per gram
H	hyperemic multiplier

Key conceptual constraint A Glagov outer-wall contour changes CT-FFR only when it is placed in the correct term of the reduced-order pressure-drop model. The actual lumen must remain in the distributed viscous integral — blood flows through the observed lumen, not the reconstructed reference. The reference geometry belongs in the contraction term, where pressure loss depends on how strongly the lumen contracts relative to the healthy reference vessel. This distinction is not pedantic: placing the outer-wall correction in the viscous term would incorrectly reduce the distributed resistance estimate in the stenotic segment. The proposal adopts this constraint explicitly as a design commitment across all four model variants, and tests whether it produces a clinically meaningful improvement in diagnostic accuracy against invasive FFR.

Model equations Distributed viscous term

$$\Delta P_{\text{visc}} = K_v \mu Q \int_0^L \frac{ds}{A_{\text{lum}}(s)^2}$$

This lumen-based term follows standard reduced-order stenosis modeling because viscous loss is determined by the area through which blood actually flows²⁶. Exact circular-tube Poiseuille theory gives the same $\mu Q \int ds/A^2$ structure, with area-form coefficient 8π ; the present proposal adopts **primary** $K_v = 32$ as a Young-Tsai-style reduced-order closure to absorb non-ideal stenosis geometry and profile effects rather than treating it as a first-principles identity. Sensitivity analyses will bracket this choice by varying K_v from the exact circular coefficient 8π up to the adopted reduced-order convention 32. Note:²⁷ extend the two-term form with an explicit turbulence term because the basic K_v closure is insufficient at intermediate-to-severe stenoses; this extension is deferred to post-validation development. To preserve interpretability of the 2×2 ablation, K_v **will be fixed prospectively before**

primary endpoint evaluation, identical across all four model variants.

Reference-dependent contraction term

$$\Delta P_{\text{sep}} = K_i \frac{\rho}{2} Q^2 \left(\frac{1}{A_{\text{MLA}}} - \frac{1}{A_{\text{ref,MLA}}} \right)^2$$

This is the point at which Glagov reference geometry enters the hemodynamics: as a reference-dependent contraction loss, not as a replacement for the observed lumen in the viscous integral²⁶. The squared inverse-area difference is the reduced-order form of the local velocity jump, where $\rho = 1050 \text{ kg/m}^3$ denotes blood density, while K_i is treated as an adopted loss coefficient rather than as a first-principles constant. Primary analyses will use $K_i = 1.2$ as a pragmatic reduced-order default, with sensitivity analysis over $[1.0, 1.5]$. As with K_v , K_i **will be fixed a priori across the full 2×2 model family** so that performance differences are attributed to geometry and flow allocation rather than to model-specific parameter tuning.

Here, $A_{\text{ref,MLA}}$ is taken from the reconstructed local reference profile at the MLA position. In tapering vessels or lesions with distal recontraction, this remains a reduced-order approximation to the classical Young-Tsai setting rather than an exact immediate-expansion recovery²⁶.

Total pressure drop

$$\Delta P = \Delta P_{\text{visc}} + \Delta P_{\text{sep}}$$

CT-FFR

$$\text{CT-FFR} = 1 - \frac{\Delta P}{P_a}$$

This follows directly from $\text{FFR} = P_d/P_a$ with $P_d = P_a - \Delta P$ once the total reduced-order pressure drop has been specified. For development-phase analyses, P_a will be set to a **nominal aortic pressure of 100 mmHg** — used as a population-independent planning constant consistent with the reduced-order CT-FFR convention, not as a patient-specific measurement — with sensitivity analysis over **90–110 mmHg** spanning the measured mean hyperemic aortic pressures in the FAME, NXT, and DISCOVER-FLOW populations^{3,28,29}. The critical modeling point is that Glagov reference geometry changes the contraction term, not the lumen integral.

Diameter-based comparator For the baseline diameter-flow models, vessel demand is estimated from proximal caliber:

$$Q_{\text{rest},v} = Q_0 \left(\frac{D_{\text{prox},v}}{3.0 \text{ mm}} \right)^{2.7}$$

$$Q_{\text{hyp},v} = H Q_{\text{rest},v}$$

with benchmark normalization $Q_0 = 120$ mL/min at $D_0 = 3.0$ mm and primary $H = 3.5^{12,30}$. This comparator matches the boundary-condition convention of prior CT-FFR studies^{11,31} and is treated as a benchmark, not as the preferred physiological formulation.

Mass-based flow formulation The primary vessel-demand model is:

$$M_v = f_v M_{\text{cal}}$$

$$Q_{\text{rest},v} = q_{\text{rest}} M_v$$

$$Q_{\text{hyp},v} = H Q_{\text{rest},v}$$

with $q_{\text{rest}} \approx 1.0$ mL/min/g¹², f_v from dominance-adjusted anatomical territory fractions (table below), primary $H = 3.5^{12}$ with **population-level sensitivity range** [2.5, 4.0] (spanning the ± 1 SD range around the healthy-volunteer mean). This range is distinct from disease-stratified H values: the population range treats all patients identically and quantifies measurement and physiological variability, whereas disease-stratified values (nominally $H = 2.8$ for intermediate CAD per³², $H = 2.5$ for severe CAD as a modeled conservative reduction) assign subgroup-specific hyperaemia and will be evaluated as secondary sensitivity analyses; they are not embedded in the primary model. An allometric sensitivity branch $Q \propto M_v^\gamma$ with $\gamma \in [0.75, 1.0]$ normalized to the primary linear model at the reference mass will also be evaluated; the primary normalization fixes $M_{\text{ref,cal}} = 250$ g so that $M_{\text{ref},v} = f_v M_{\text{ref,cal}}$, $\gamma = 1.0$ is the primary choice (constant perfusion per gram, consistent with¹²), and $\gamma = 0.75$ is the cross-species allometric alternative^{13,33}.

The following figure visualizes the physiological motivation for this formulation.

Territory fractions Because measured perfusion territories will not be available for most development-phase cases, the primary model uses anatomical fractions with coronary-dominance adjustment:

Dominance	LAD	LCx	RCA
Right-dominant	0.42	0.30	0.28
Left-dominant	0.42	0.38	0.20
Co-dominant	0.42	0.33	0.25

Fractions follow dominance-adjusted anatomical data³⁴; the LAD fraction is stable across dominance patterns while LCx/RCA redistribution absorbs dominance variation. CT-derived mass tracks invasive

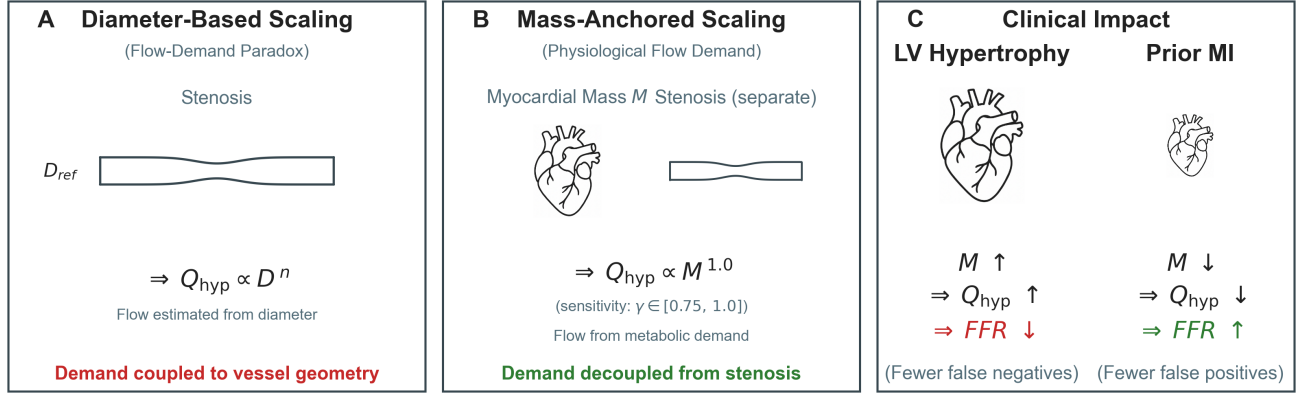


Figure 5: **Mass-anchored flow estimation.** Flow demand is determined by myocardial tissue mass, not vessel geometry. (A) Conventional diameter-based flow scaling ties demand to observed diameter and creates the flow-demand paradox. (B) Mass-anchored scaling uses primary linear mass allocation ($\gamma = 1.0$) with sublinear allometric sensitivity ($\gamma = 0.75$) to decouple demand from stenosis geometry. (C) Clinical scenarios: left-ventricular hypertrophy (increased mass $\rightarrow$ higher flow demand $\rightarrow$ lower FFR) and prior myocardial infarction (reduced viable mass $\rightarrow$ lower flow demand $\rightarrow$ higher FFR).

territorial perfusion³⁵, so the mass-based arm tests whether patient-specific demand improves over proximal caliber scaling. Fixed-fraction sensitivity and an exploratory Voronoi-style territorial assignment will be evaluated secondarily. For the lumen-only geometry arm, A_{ref} is the monotone lumen envelope, so all four model variants share identical pressure-drop equations and differ only in $A_{ref,MLA}$ and $Q_{hyp,v}$.

Glagov reference reconstruction For cases with usable outer-wall data:

$$T_{healthy} = \text{median}(r_{outer} - r_{lum})$$

evaluated over trusted proximal and distal reference slices (clipped to **0.40–0.80 mm**^{36–38}), giving:

$$r_{ref,G}(s) = r_{outer}(s) - T_{healthy}, \quad r_{ref,G}(s) \geq 1.05 r_{lum}(s)$$

with monotonic smoothing. The floor $\geq 1.05 r_{lum}$ prevents implausible inversions but does not cap Glagov correction in strongly remodelled segments.

Eligibility and fallback hierarchy. Glagov mode is used only when the trusted outer-wall interval covers the MLA and extends ≥ 5 mm or one vessel diameter (whichever is larger) in both directions, with $\leq 10\%$ severely artefacted slices. Cases failing these criteria revert to the lumen-envelope comparator. Where the local wall estimate is unstable but the interval is otherwise eligible, a fallback prior $T_{healthy} = \beta \tilde{r}_{lum}$, $\beta \approx 0.18$ ³⁸, is applied; deeper population priors trigger reversion or exclusion from primary G-CT-FFR / MG-CT-FFR analyses. An exploratory remodeling index $RI_{Glagov} = A_{outer}/A_{ref,G}$ at the MLA will be retained only as a Level-0-supported descriptive ratio in the current cycle; no positive/intermediate/constrictive threshold claim is locked.

2.3.1 Work programme

WP	Months	Main output	Deliverable
WP1	1–4	FFR-linked cohort assembly and geometry QC	QC-locked analyzable cohort table plus baseline L-CT-FFR code and validation memo
WP2	5–9	2×2 model implementation and first invasive validation	Internal model-comparison report with bootstrap CIs and updated methods package
WP3	10–12	Hierarchical reference-reconstruction strategy for unreliable outer-wall data	Locked fallback-mode definitions, MLA-support rules, and Glagov-eligibility reporting package
WP4	13–16	Robustness, reader agreement, and segmentation reproducibility	Robustness/QC report and draft methods manuscript material
WP5	17–24	Expanded harmonized cohort processing and exploratory prognosis	Exploratory cohort analysis, software release package, and submission-ready manuscripts

WP3 is the explicit fallback-hierarchy work package — a self-contained deliverable, not a hidden sub-task inside generic robustness analyses. Operationally, WP3 outer-wall eligibility classification and fallback rule development runs in parallel with WP2 model validation: the two streams share the same geometry input but answer different questions (WP2: which model wins on analyzable cases; WP3: what fraction of cases are analyzable at each evidence level). The WP2 → WP3 transition criterion therefore marks when the core model is locked rather than when WP3 work begins.

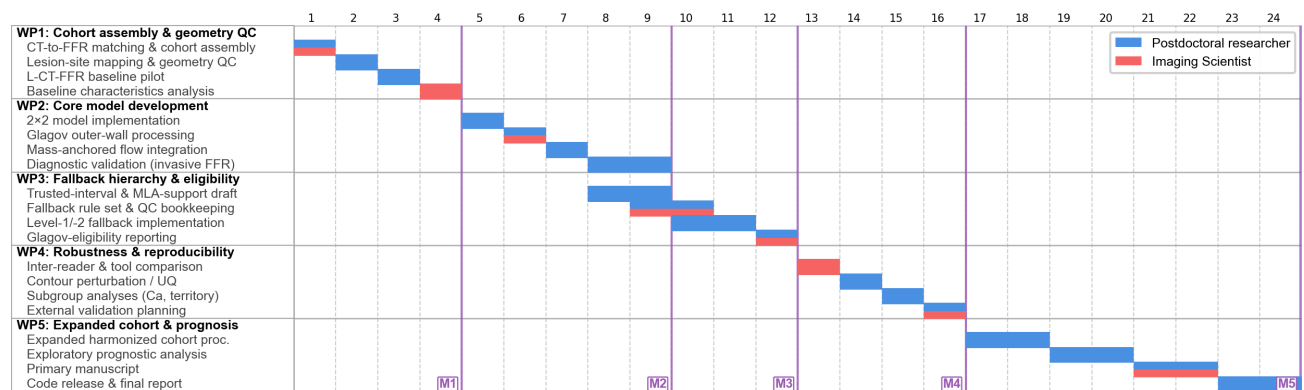


Figure 6: **Project timeline (24 months, five work packages)**. Bars are colour-coded by personnel role: blue = postdoctoral researcher, red = imaging scientist, split bars = shared effort. Milestones M1–M5 mark the end-of-WP deliverables (vertical lines).

Work package transitions will be judged against explicit operational criteria:

- **WP1 → WP2:** matched CTA-FFR inventory locked and pilot L-CT-FFR processing success at or above **80%** in the first 30 Stage-A vessels undergoing QC, where “processing success” means the locked lumen-only baseline workflow returns either a numeric L-CT-FFR value or an explicit analyzability classification without unresolved geometry-ordering failure;
- **WP2 → WP3:** all four fixed-parameter models executable on the strict four-arm complete-case primary subset with no unresolved geometry-ordering failures in retained vessels;
- **WP3 → WP4:** fallback hierarchy locked, with Glagov-eligibility rate and fallback frequencies available as pre-specified reportable quantities;
- **WP4 → WP5:** reader/tool robustness analyses and perturbation-based uncertainty quantification complete, with harmonization judged sufficient for exploratory patient-level modelling.

The staged structure is also the contingency plan. If analyzable yield or outer-wall completeness is lower than anticipated, the minimum viable deliverable remains a validated lumen-only benchmark against invasive FFR, a pre-specified 2×2 geometry-versus-demand comparison on the analyzable subset, and a locked fallback/QC framework for unreliable outer-wall cases.

2.3.2 Primary endpoint The **primary endpoint** is per-vessel diagnostic agreement with invasive FFR in the FFR-linked development cohort.

The **primary comparison metric** is AUC for invasive $\text{FFR} \leq 0.80^{1,3}$ on the strict four-arm complete-case paired target-vessel subset. Supportive diagnostic metrics are:

- mean absolute error versus invasive FFR;
- Bland-Altman bias and limits of agreement;
- calibration slope and intercept;
- threshold agreement at 0.80.

The **primary unit of analysis** is the paired target-vessel comparison defined by one invasive FFR measurement and its matched CTA vessel. If a patient contributes more than one analyzable vessel, confidence intervals will be obtained by **patient-clustered bootstrap**.

The primary model-comparison procedure will use **paired DeLong contrasts for AUC** with **Holm correction** across the three planned superiority-style comparisons against **L-CT-FFR**. A repeated-measures regression with **geometry**, **flow-allocation**, and **geometry × flow** interaction terms will be reported as a supportive secondary inferential analysis to estimate additive versus non-additive gains directly, but it is not part of the primary Holm-controlled family. The study is sized for development-phase detection of moderate paired gains in the broader matched-vessel planning pool (nominal $\Delta\text{AUC} \approx 0.09\text{--}0.10$); the final threshold on the strict complete-case primary subset will be larger, and differences below that range will be treated as estimation signals rather than definitive superiority findings.

2.3.3 Secondary and exploratory analyses Secondary:

- segmentation robustness;
- inter-reader agreement;

- Glagov-eligibility rate, fallback-level frequencies, and clean-Level-0 versus fallback-supported subset performance;
- exploratory RI_Glagov distribution and its relation to the L-versus-G model difference in Level-0 cases;
- perturbation-based uncertainty quantification for lumen, outer-wall, and myocardial-mass inputs in representative vessels;
- contribution of geometry vs demand corrections.

Exploratory:

- patient-level outcome associations in a larger harmonized CT subset.

2.4 Handling of research data

All data originate from DISCHARGE and will be processed within the applicant institution. A central design commitment is that the existing vessel segmentations are not assumed sufficient for immediate whole-cohort flow derivation.

The data workflow therefore includes:

- segmentation harmonization;
- vessel-level quality control;
- lesion-to-FFR matching;
- derivation of analyzable vessel segments;
- release of code and derived variables where allowed.

2.4.1 Data basis, governance, and legal framework DISCHARGE data governance is established through the applicant institution as coordinating centre of the DISCHARGE trial. The analysis uses pseudonymized trial imaging and follow-up data under the existing governance framework for secondary scientific evaluation. Processing will occur entirely within secure institutional infrastructure; no external transfer of individual-level raw DISCHARGE data is required for the project. The relevant ethical basis is the existing DISCHARGE approval at Charité (EA4/064/11) together with the original multicentre consent structure for secondary analysis.

2.4.2 Data quality and protection The project benefits from standardized DISCHARGE acquisition and adjudication procedures, central imaging review, and existing vessel-segmentation work from the Medis-based SPP 2177 effort. The remaining technical challenge is therefore not basic data existence, but harmonization, geometry QC, and analyzability filtering. Raw DISCHARGE images remain protected. Derived model outputs and analysis scripts will be shared only within institutional, legal, and ethical constraints.

2.4.3 Open science and reproducibility We commit to open and reproducible methods wherever patient protection permits:

- code for the reduced-order CT-FFR pipeline will be released on GitHub under an open licence as modular Python components built around standard scientific libraries;
- documented workflows for sensitivity analyses, fallback/QC summaries, and perturbation-based uncertainty quantification will be released alongside the core code;
- derived de-identified variables will be shared where governance permits;
- the main analysis plan will be pre-specified before final endpoint evaluation;
- publications will be submitted with open-access options where institutionally available.

2.5 Relevance of sex, gender and/or diversity

Sex-specific physiology is directly relevant to this proposal. Reference LV myocardial mass differs substantially by sex: population-level CMR normative studies give approximately **95 g** (female) and **140 g** (male)^{39,40}. In the current revision cycle, the primary allometric anchor remains the cohort-wide constant $M_{\text{ref,cal}} = 250 \text{ g}$; the sex-specific values are used instead as plausibility context for CT-derived mass outliers, as a sensitivity check on calibration residuals, and as a stratification variable for exploratory subgroup analyses. Coronary geometry is also sexually dimorphic, with smaller vessel calibers in women likely amplifying both the Glagov bias and the diameter-flow paradox⁴¹.

Formal subgroup inference on sex will remain exploratory unless the analyzable FFR-linked sample supports it (practical rule: ≥ 10 events or comparisons per stratum). Descriptive reporting of sex-stratified model performance, LV mass distributions, and Glagov-eligibility rates is planned regardless of subsample size.

3 Project- and subject-related list of publications

The works cited in sections 1 and 2 are generated automatically below from the shared bibliography. In the submission version, up to ten of the applicant's own project-related works can still be highlighted manually if required by the DFG template.

1. Pijls NHJ, De Bruyne B, Peels K, et al. Measurement of fractional flow reserve to assess the functional severity of coronary-artery stenoses. *New England Journal of Medicine*. 1996;334(26):1703–8. doi:10.1056/NEJM199606273342604
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4 Supplementary information on the research context

4.1 Ethical and/or legal aspects of the project

4.1.1 General ethical aspects This is a **low-risk retrospective analysis** of existing DISCHARGE trial data. The project involves:

- secondary analysis of previously collected imaging data (CT angiography, myocardial segmentations);
- no additional patient contact, interventions, or data collection;
- no experimental treatments or diagnostic procedures;
- computational processing of pseudonymised imaging data.

The research poses no physical risk to participants. The primary ethical consideration is data privacy, which is addressed through existing DISCHARGE data governance structures.

4.1.2 Descriptions of proposed investigations on humans, human materials or identifiable data

This project uses DISCHARGE trial imaging and clinical follow-up data from human participants. Access is justified because the applicant institution (Charité) is the DISCHARGE coordinating centre; all project-level analyses are performed on pseudonymised datasets within the existing governance framework.

Characteristic	DISCHARGE
Participants	3,561 patients; stable chest pain; intermediate CAD probability
Study type	Randomised multicentre (26 centres, 16 countries)
CT scanners	>10 models (Canon, GE, Siemens, Philips)
Primary endpoint	MACE (CV death, non-fatal MI, non-fatal stroke)
MACE events (CT arm)	38 events at 3.5-year median follow-up
Recruitment	October 2015 – January 2019
Extended follow-up	target ~8–10 years total follow-up; operational date to refresh at submission

Table 7: Overview and clinical endpoints of the DISCHARGE trial.

Medis Segmentation Reader Breakdown (SPP 2177): Reader 1 analysed 6,639 vessel segments in 1,653 patients; Reader 2 analysed 3,421 segments in 740 patients. In 709 patients, both readers provided segmentations, enabling inter-observer reproducibility analyses. The combined unique-patient total is 1,684 (1,653 Reader-1 patients plus 31 patients analysed exclusively by Reader 2). Inner-lumen and outer-wall contours are exported as dense 3D point clouds sampled along straightened multiplanar reformations (MPRs) and at individual cross-sections.

DISCHARGE Ethics Approval: Charité—Universitätsmedizin Berlin, Ethics Committee approval

EA4/064/11. The original DISCHARGE study received ethics approval from all 26 participating centres across 16 European countries.

Informed Consent: All DISCHARGE participants (n=3,561) provided written informed consent prior to enrolment for:

- collection and storage of imaging data;
- secondary analysis for cardiovascular research;
- publication of anonymised research findings;
- long-term follow-up and retrospective data use.

Data Access Procedures: The applicant institution (Charité) is the DISCHARGE coordinating centre. Data access for this project is governed by the existing DISCHARGE data governance framework established during the original trial. All processing occurs on-site within the coordinating centre; no external data sharing or transfer is required. No additional ethics approval is needed for this secondary analysis of existing trial data.

Pseudonymisation and Data Protection:

- all data are pseudonymised at source using study-specific patient identifiers;
- original patient identifiers are stored separately and are not accessible for this project;
- processing occurs exclusively within Charité's secure IT infrastructure;
- no individual-level data will leave the institution;
- only aggregate results and de-identified derived variables will be published;
- full compliance with GDPR (EU 2016/679) and the German Federal Data Protection Act (BDSG).

4.1.3 Descriptions of proposed investigations involving experiments on animals N/A—This project involves no animal research.

4.1.4 Descriptions of projects involving genetic resources (or associated traditional knowledge) from a foreign country N/A—This project does not involve genetic resources or traditional knowledge.

4.1.5 Explanations regarding any possible safety-related aspects

4.1.5.1 “Dual Use Research of Concern”; foreign trade law N/A—This project develops diagnostic algorithms for medical imaging. The research has no dual-use potential and does not involve controlled technologies or materials.

4.1.5.2 Risks in international cooperation N/A—The core project involves no international data transfer. All analyses are performed on pseudonymised DISCHARGE data within Charité's institutional infrastructure.

4.1.6 Considerations on aspects of ecological sustainability in the planning and implementation of the project

Computational sustainability: This project has a low environmental footprint. The reduced-order modelling approach requires only seconds per case (versus hours for 3D CFD); total project compute time is expected to be on the order of ~30 hours on a single workstation, or <3 hours on a 16-core cluster backup, with negligible energy consumption. No new hardware procurement is required.

Sustainability measures:

- use of existing institutional computing infrastructure;
- efficient algorithms enabling rapid sensitivity analysis without large-scale cluster use;
- open-source code release enables reuse and avoids duplication of effort;
- virtual meetings preferred for routine collaboration.

4.2 Employment status information

- **Dr. rer. med. Steffen Lukas**, Charité Universitätsmedizin Berlin: PI position permanently funded by the institution.

4.3 First-time proposal data

This is not **Steffen Lukas's** first DFG proposal; the first-time proposal information is therefore not applicable.

4.4 Composition of the project group

The following individuals will contribute to the project without salary support from project funds:

They participate through existing institutional roles or other project-external support; no additional salary line is requested for them in this proposal.

- **Medical team:** Federico Biavati MD; Moritz Gross MD; PD Dr. med. Florian Michallek; Dr. med. Ann-Christine Stahl; Sotirios Tsogias MD
- **Non-medical team:** Mahmoud Mohamed MSc; Kenrick Schulze MSc; Dr. rer. nat. Maria Bosserdt; Bernhard Föllmer MSc; Dr. rer. med. Steffen Lukas

4.5 Researchers in Germany with whom you have agreed to cooperate on this project

- **DISCHARGE Consortium** (Charité and participating German centres): access to imaging data, clinical outcomes, and patient characteristics; established data governance and quality-control procedures; methodological input on CT imaging protocols and segmentation standards.
- **German Cardiac CT Working Group:** expert review of coronary CT methodology, clinical relevance, and translational positioning of the reduced-order CT-FFR framework.
- **Dr. Christoph von Tycowicz**, Zuse Institute Berlin (ZIB), Visual and Data-centric Computing, Geometric Data Analysis and Processing; methodological exchange on geometric data analysis

and processing for robust vessel-geometry handling and computational pipeline design. Tel. +49 30 84185-350.

4.6 Researchers abroad with whom you have agreed to cooperate on this project

No formal collaboration with researchers abroad is planned for the core project period. The primary development and validation work relies entirely on DISCHARGE data processed within the applicant institution. External validation against an independent cohort is an explicitly post-grant objective: subject to successful internal DISCHARGE validation and data-access agreements, planned targets include the CTA-imaged SCOT-HEART cohort (Prof. David E. Newby, Edinburgh) and the PROMISE CT arm, both of which contain coronary CT imaging and clinical follow-up compatible with the proposed analysis framework^{42,43}.

4.7 Researchers with whom you have collaborated scientifically within the past three years

See attached CVs for full publication lists and recent scientific collaborations.

4.8 Project-relevant cooperation with commercial enterprises

None. This project does not involve industry collaboration or funding. The open-source release strategy explicitly avoids proprietary restrictions.

Segmentation software: Medis QAngio CT workbench (Medis Medical Imaging Systems, Leiden, Netherlands) was used for DISCHARGE plaque annotations under the completed SPP 2177 project. This is a standard commercial tool; no special collaboration exists.

4.9 Project-relevant participation in commercial enterprises

None. The applicant has no financial interests in CT-FFR companies or related enterprises.

4.10 Scientific equipment

Existing infrastructure (no procurement required):

The project relies exclusively on existing institutional computing resources. Charité provides a secure high-performance computing (HPC) environment for shared use and on-site workstations for plaque analysis and prototyping. Central data storage is available via the institutional server infrastructure.

For the planned reduced-order modelling, paired model comparisons, sensitivity analyses, and perturbation-based uncertainty propagation, a single 16-core workstation with 64 GB RAM and 2 TB SSD is sufficient; the institutional HPC cluster serves as backup for parallel runs.

- **Primary workstation:** 16-core CPU, 64 GB RAM, 2 TB SSD (available in PI's lab; sufficient for all analyses)
- **HPC cluster (backup):** multi-node, SLURM-managed (institutional access; available if needed)
- **Storage:** existing institutional allocation; estimated project data volume ~60 GB

Software and analysis tools: Primary analyses use open-source software (Python, NumPy, SciPy) and existing in-house tools for calcium scoring and coronary segmentation. No additional proprietary dependencies are required for the core reduced-order workflow.

Compute requirements: Point-estimate CT-FFR runs, paired model comparisons, sensitivity analyses, and perturbation-based uncertainty propagation are expected to require roughly ~30 hours on a single workstation and <3 hours on a 16-core cluster. Computational resources are **not a bottleneck**.

4.11 Other submissions

None. This project has not been submitted to other funding agencies.

4.12 Other information

Use of AI Tools: This proposal's text clarity and readability were improved with AI-assisted language polishing, including Anthropic Claude. AI tools were not used for hypothesis formulation, data analysis, or image generation.

Open Science Commitment:

- all core algorithm code will be released on GitHub under the **MIT licence**;
- the primary analysis plan will be pre-registered on **OSF** before final primary-endpoint evaluation;
- de-identified derived variable tables and QC metadata will be deposited on **Zenodo** under **CC-BY-4.0** where governance permits;
- publications will be pursued with open-access options where institutionally available.

Auditability: Compared with opaque machine-learning surrogates, the proposed low-parameter reduced-order model is easier to audit, document, and supervise. This transparency is a methodological advantage even within a research-only setting.

Post-grant pathway: Independent-cohort validation, regulatory qualification, and clinical deployment are explicitly post-grant objectives. If the DISCHARGE development programme succeeds, the next translational steps would include external validation and later consideration of the relevant European and U.S. regulatory pathways.

Risk management and minimum viable deliverable: The major operational risks are lower-than-expected analyzable yield in the invasive-FFR subset after matching and QC, variable outer-wall quality in calcified or truncated segments, and slower-than-expected harmonization of broader cohort geometry. These risks do not invalidate the project but affect the scope of what can be claimed. The minimum viable scientific deliverable is: (1) a validated lumen-only development benchmark against invasive FFR; (2) a pre-specified 2×2 geometry-versus-demand comparison on the analyzable subset; (3) a documented segmentation and QC framework with explicit fallback logic for unreliable outer-wall cases.

Scope limitation — serial and tandem lesions: The development-phase reduced-order model is centred on a single dominant minimum lumen area (MLA) per vessel span. Vessels with serial or

tandem lesions within one analyzed span are not handled in the primary model and should be deferred to a dedicated multi-lesion extension in later work⁴⁴.

Project outputs: Beyond journal manuscripts, the project is intended to deliver a documented reduced-order CT-FFR codebase, reproducible validation and sensitivity-analysis workflows, de-identified derived variable tables where governance permits, and a methods-oriented software package suitable for later extension. Open dissemination is a design goal but remains subordinate to DISCHARGE data-protection constraints.

5 Requested modules

5.1 Basic Module

5.1.1 Funding for Staff Personnel overview:

Role	FTE	Hours/mo	Total (24 mo)
Postdoc (computational)	1.0	160	3,840
Imaging scientist	0.2	32	768
PI (supervision; institutional contribution)	0.1	16	384

Table 8: Personnel effort allocation.

Total project effort: approximately 4,992 person-hours over 24 months.

Requested budget summary: 215,540 EUR total = 209,040 EUR personnel + 5,000 EUR travel + 1,500 EUR publication expenses.

Budget calculation (DFG Personnel Rates 2026):

The DFG approves funding for staff in the form of standard amounts based on gross employer costs, including employer contributions for social security and annual special payments.

Position	Pay Grade	Rate (EUR/yr)	Duration	Amount
Postdoc (computational)	Postdoctoral researcher	88,200	24 months	176,400
Imaging scientist (20%)	Doctoral researcher	16,320	24 months	32,640
Personnel Subtotal				209,040

Table 9: Personnel budget calculation (DFG standard rates 2026).

Justification:

- **Postdoc (1.0 FTE, postdoctoral researcher rate):** The staged 2×2 model development, cohort assembly, invasive-FFR validation, sensitivity analyses, and primary manuscript preparation require a full-time researcher with PhD-level expertise in biomedical engineering, medical physics, or computational cardiovascular science. The position is funded at the DFG postdoctoral researcher standard amount for 2026 (88,200 EUR/year).

- **Imaging scientist (0.2 FTE, doctoral researcher rate):** Design and methodological coordination of the segmentation QC strategy, planning and analysis of the inter-observer reproducibility substudy, formalisation of outer-wall fallback and exclusion rules, and preparation of methodological figures require specialised cardiac CT expertise. The 0.2 FTE allocation (~32 hours/month) is funded at the DFG doctoral researcher standard amount for 2026 (81,600 EUR/year, prorated to 0.2 FTE) and covers study design, protocol development, and supervision of QC procedures, while routine day-to-day QC tasks are performed by existing DISCHARGE imaging staff within their institutional duties.

5.1.2 Direct Project Costs

5.1.2.1 Equipment up to €10,000, Software and Consumables Existing computing infrastructure (Section 4.10) is sufficient. All required software packages are covered by existing institutional agreements. No additional equipment or software funds are requested in this module. Primary analysis relies on open-source software (Python, NumPy, SciPy) and existing institutional computing infrastructure.

Any required commercial packages are already licensed via institutional agreements; no hidden software costs are expected within the requested budget.

5.1.2.2 Travel Conference dissemination budget:

Conference	Year	Purpose	Amount
ECR, Vienna	1	Preliminary results; radiology/cardiac CT feedback	1,500
SCCT Annual Meeting	2	CT-FFR validation; community networking	2,000
MICCAI or MIDL	2	Algorithmic contributions; open-source release	1,500
Subtotal			5,000

Table 10: Travel budget for conference dissemination.

Justification: Conference presentations are essential for scientific dissemination, methodological scrutiny, and community feedback. The selected venues cover (i) radiology/cardiac CT readers (ECR), (ii) the dedicated cardiac CT/CT-FFR community (SCCT), and (iii) the technical medical imaging and computational methods community (MICCAI/MIDL). The budget is a conservative estimate for one attendee per meeting and includes registration, economy travel (rail/flight), accommodation (typically 3–5 nights), and per diem in line with institutional travel regulations.

5.1.2.3 Visiting Researchers (excluding Mercator Fellows) N/A—No visiting researchers planned.

5.1.2.4 Expenses for Laboratory Animals N/A—This project involves no animal research.

5.1.2.5 Other Costs N/A—No other direct costs anticipated.

5.1.2.6 Project-related Publication Expenses Project results will be disseminated via quality-assured open-access or hybrid journals with green-open-access fallback. Two peer-reviewed publications are planned: one development-validation manuscript and one methods manuscript. In addition, manuscripts will be posted to preprint repositories such as arXiv and bioRxiv where journal policy permits.

Publication	Target venue class	Amount
Development-validation manuscript	Open-access journal with institutional OA coverage where possible	750
Methods manuscript	Technical imaging/methods journal with institutional OA coverage where possible	750
Total		1,500
pub- li- ca- tion re- serve		

Table 11: Publication expenses.

Note: This amount is a contingency reserve for publication costs not already covered by institutional open-access agreements or green-open-access routes. Final journal selection will preferentially use agreement-covered venues where possible.

5.1.3 Instrumentation

5.1.3.1 Equipment Exceeding €10,000 **None required.** All analyses can be performed on existing institutional infrastructure (Section 4.10).

5.1.3.2 Major Instrumentation Exceeding €50,000 **None required.** No major equipment procurement is necessary for this project.

5.2 Module Temporary Position for Principal Investigator

N/A—PI position is permanently funded by the institution.

5.3 Module Replacements

N/A—No replacement funding requested.

5.4 Module Temporary Substitutes for Clinicians

N/A—No clinician substitute required.

5.5 Module Mercator Fellows

N/A—No Mercator Fellows planned.

5.6 Module Project-Specific Workshops

N/A—No workshop funding requested.

5.7 Module Public Relations

N/A—No public relations funding requested. Dissemination will occur through standard academic channels (publications, conferences, GitHub).

5.8 Module Standard Allowance for Gender Equality Measures

N/A—No standard allowance for gender equality measures requested.